

ME 356: Principles of Energy Conversion

Course Code	ME 356
Course Name	Principles of Energy Conversion
L-T-P-C	3-1-0-4
Prerequisites	Thermodynamics
Intended for	B.Tech.
Distribution	Elective
One-time approval	3rd Ad-hoc Senate (Nov 2012)

Course Contents

- Introduction to power system and technologies. Demand variation and forecasting. Grid features. Siting and costing.
- Diesel generators: system, equipment and layout.
- Fossil-fuelled steam power plants: boiler and accessories. Turbine and accessories, feed cycle equipment, generator.
- Combined cycle power plants: gas turbine, heat recovery boiler.
- **Nuclear power:** nuclear reactions, fuel, moderator and coolant. Neutron life cycle. Light water, heavy water, gas cooled and fast reactors.
- Hydroelectric plants: features and siting, Pelton, Francis, Kaplan and propeller turbines construction, mini and micro turbines.
- **Renewable energy:** solar, geothermal, wind, biomass, ocean, fuel cells, unique features of decentralized systems. Co-generation systems. Environmental issues, sustainability and future scenarios.

Text Books

1. BEI International, Hambling, P., (Ed.), **Modern Power Station Practice: Nuclear Turbines, and Associated Plant**, Pergamon Press, 1992.
2. Drbal, L. F., Boston, P. G., Westra, K. L., Black and Veatch, **Power Plant Engineering**, Kluwer Academic, 1995.
3. Elliott, T. C., Chen, K., and Swanekamp, R., **Standard Handbook of Power Plant Engineering**, 2nd edition, McGraw-Hill Professional, 1997
4. El-Wakil, M. M., **Power Plant Technology**, McGraw-Hill, 1984.
5. Jog, M., **Hydro-electric and Pumped Storage Plants**, John Wiley, 1989.
6. Fritz, J. J., **Small and Mini Hydropower Systems**, McGraw-Hill, 1984.
7. Central Board for Irrigation and Power (CPIB), India, Design and Construction Features of Selected Dams in India, 1983.
8. Borbely, Anne-Marie, and Kreider, Jan J., (Eds.), **Distributed Generation: The Paradigm for the New Millennium**, CRC Press, 2003.
9. Larminie, J., and Dicks, A., **Fuel Cell Systems Explained**, John Wiley, 2003.
10. Vielstich, W., Lamm, A., and Gasteiger, H., **Handbook of Fuel Cells: Fundamentals, Technology, Applications**, John Wiley, 2003.
11. O'Hayre, R., Cha, S. -W., Colella, W., and Prinz, F. B., **Fuel Cell Fundamentals**, Wiley, 2005.
12. Appleby, A. J., and Foulkes, F. R., **Fuel Cell Handbook**, van Nostrand Reinhold, 1996.
13. Harrison, R., Hau, E., and Snel, H., **Large Wind Turbines: Design and Economics**, John Wiley, 2001.
14. Bejan, Adrian, **Advanced Engineering Thermodynamics**, Wiley, 1997.
15. Patents and catalogues related to various equipment.